Homework 5

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1 Inverse Comb Filter

$$y_t = x_t - R^L x_{t-L},\ R = 0.999,\ L = 6$$

1.1 Transfer Function $\mathcal{H}(z)$

$$\begin{aligned} Y &= X(1 - R^L z^{-L}) \\ \mathcal{H}(z) &= 1 - 0.999^6 z^{-6} \end{aligned}$$

1.2 Frequency Response $H(\omega)$

$$\begin{aligned} \mathcal{H}(e^{i\omega}) &= 1 - R^L e^{-iL\omega} \\ H(\omega) &= 1 - 0.999^6 e^{-i6\omega} \end{aligned}$$

1.3 Magnitude Response $|H(\omega)|$

$$\begin{aligned} H(\omega) &= e^{-iL\omega}(e^{iL\omega} - R^L) \\ |H(\omega)| &= |e^{iL\omega} - R^L| \\ |H(\omega)| &= |\cos(L\omega) + i\sin(L\omega) - R^L| \\ |H(\omega)| &= \sqrt{(\cos(L\omega) - R^L)^2 + \sin^2(L\omega)} \\ |H(\omega)| &= \sqrt{\cos^2(L\omega) - 2R^L\cos(L\omega) + R^{2L} + \sin^2(L\omega)} \\ |H(\omega)| &= \sqrt{1 - 2R^L\cos(L\omega) + R^{2L}} \\ |H(\omega)| &= \sqrt{(1 + 0.999^{12}) - 2(0.999)^6\cos(6\omega)} \end{aligned}$$

1.4 Maxima and Minima of $|H(\omega)|$

To find the maxima, one must find all values of ω where $\cos(6\omega) = -1$. Therefore, the following maxima can be found:

$$\omega = \frac{k\pi}{L},\ k = 2n + 1,\ n \in \mathbb{Z}$$

Similar to the maxima, minima are all values of ω where $\cos(6\omega) = 1$. Therefore, the following minima can be found:

$$\omega = \frac{k\pi}{L},\ k = 2n,\ n \in \mathbb{Z}$$

To convert these to sampling rate, use identity $\omega = 2\pi\frac{f}{f_s}$. By substitution, the following is derived:

$$f_{\max} = \frac{kf_s}{2L},\ k = 2n+1$$
$$f_{\min} = \frac{kf_s}{2L},\ k = 2n$$

Where $n \in \mathbb{Z}$
This results in the following points:

$$\max(|H(\omega)|_{dB}) \approx \{(\frac{f_s}{12}, 5.99), (\frac{3f_s}{12}, 5.99), (\frac{5f_s}{12}, 5.99)\}$$
$$\min(|H(\omega)|_{dB}) \approx \{(0, -44.5), (\frac{2f_s}{12}, -44.5), (\frac{4f_s}{12}, -44.5), (\frac{6f_s}{12}, -44.5)\}$$

1.5 Sketch of $|H(\omega)|_{dB}$

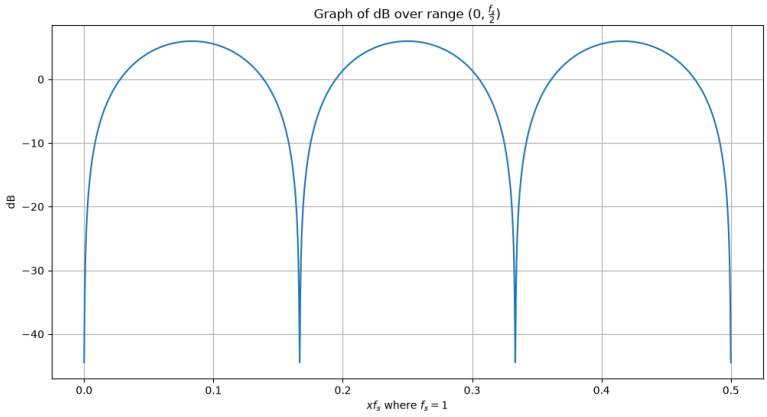


Figure 1: Magnitude Response Graph in dB

1.6 Phase response

$$
\begin{aligned}
H(\omega) &= 1 - R^L e^{-iL\omega} \\
H(\omega) &= 1 - R^L(\cos(-L\omega) + i\sin(-L\omega)) \\
H(\omega) &= (1 - R^L\cos(-L\omega)) + R^L i\sin(-L\omega) \\
\theta(\omega) &= \arctan(\frac{-R^L\sin(L\omega)}{(1 - R^L\cos(L\omega))}) \\
\theta(\omega) &= \arctan(\frac{-0.999^6\sin(6\omega)}{(1 - 0.999^6\cos(6\omega))})
\end{aligned}
$$

2 Designing a Reson Filter

One must first convert to radians per sample:

$$
\begin{aligned}
f_s &= 44100\text{Hz} \\
B_{\text{rad}} &= 2\pi\frac{B_{\text{Hz}}}{f_s}\ \text{rad/sample} \\
\psi_{\text{rad}} &= 2\pi\frac{\psi_{\text{Hz}}}{f_s}\ \text{rad/sample}
\end{aligned}
$$

Radians will be the default unit for the rest of the problem.

2.1 With $B = 2\phi,\ \cos\phi = 2 - \frac{1}{2}(R + \frac{1}{R})$

2.1.1 Deriving the pole radius formula

$$
\begin{aligned}
\phi &= \frac{10\pi}{f_s} \\
\cos\phi &= 2 - \frac{1}{2}(R + \frac{1}{R})
\end{aligned}
$$

Following the steps from class notes, one will obtain:

$$
\begin{aligned}
0 &= R^2 - 2(2 - \cos\phi)R + 1 \\
R &= \frac{2(2 - \cos(\phi)) \pm \sqrt{[-2(2 - \cos(\phi))]^2 - 4}}{2}
\end{aligned}
$$

2.1.2 Evaluating with $B = 10\text{Hz},\ \psi = 1000\text{Hz}$

Following the steps results in the filter:

$$\begin{aligned} y_t \approx & 0.00020216605481176457x_t \\ & +1.9783246236903536y_{t-1} \\ & -0.998576255974041y_{t-2} \end{aligned}$$

The impulse response is the following:

$$\begin{aligned} y_0 &= 0.00020216605481176457 \\ y_1 &= 0.00039995008430844753 \\ y_2 &= 0.00058935287793546 \end{aligned}$$

2.1.3 Evaluating with $B = 20\text{Hz},\ \psi = 5000\text{Hz}$

Following the steps results in the filter:

$$\begin{aligned} y_t \approx & 0.0018598988543063856x_t \\ & +1.511461125281618y_{t-1} \\ & -0.9971545393551153y_{t-2} \end{aligned}$$

The impulse response is the following:

$$\begin{aligned} y_0 &= 0.0018598988543063856 \\ y_1 &= 0.0028111648152399217 \\ y_2 &= 0.00239435974968163 3 \end{aligned}$$

2.2 With approximation $B = 2(1 - R)$

2.2.1 Evaluating with $B = 10\text{Hz},\ \psi = 1000\text{Hz}$

Following the steps results in the filter:

$$\begin{aligned} y_t = & 0.0002022380575210094x_t \\ & +1.978324121049828y_{t-1} \\ & -0.9985757489111912y_{t-2} \end{aligned}$$

The impulse response is the following:

$$\begin{aligned} y_0 &= 0.0002022380575210094 \\ y_1 &= 0.00040009242738807546 \\ y_2 &= 0.00058956248000382 \end{aligned}$$

2.2.2 Evaluating with $B = 20\text{Hz},\ \psi = 5000\text{Hz}$

Following the steps results in the filter:

$$\begin{aligned} y_t &= 0.0018612234956407x_t \\ &+1.5114595871848688y_{t-1} \\ &-0.9971525127908784y_{t-2} \end{aligned}$$

The impulse response is the following:

$$\begin{aligned} y_0 &= 0.0018612234956407 \\ y_1 &= 0.002813164096379871 \\ y_2 &= 0.0023960601582540676 \end{aligned}$$

2.3 Comment

They are suprisingly closer than I expected, it makes sense it's so widely used.

3 Filter Reverse Engineering

$$y_t = \sqrt{2}(0.99^t)\sin(\frac{\pi}{4}(t+1))$$

This filter can be matched with the formula:

$$y_t = R^t[\frac{\sin(\theta(t+1))}{\sin\theta}]$$

From there we can get the following:

$$\begin{aligned} R &= 0.99 \\ \theta &= \frac{\pi}{4} \end{aligned}$$

4 $R \approx e^{-\frac{B}{2}}$ Approximation Derivation

Suppose $R = 1 - \epsilon$, then one can represent $\cos\phi$ as:

$$\cos\phi = 2 - \frac{1}{2}(1 - \epsilon + (1-\epsilon)^{-1})$$

Then from the power series of $(1-\epsilon)^{-1}$:

$$\cos\phi = 2 - \frac{1}{2}(1 - \epsilon + 1 + \epsilon + \epsilon^2 + \dots)$$

After simplifying a bit:

$$\cos\phi = 1 - \frac{1}{2}\epsilon^2 + \dots$$

Using the expansion of $\cos\phi$:

$$\cos\phi = 1 - \frac{1}{2}\phi^2 + \dots$$

From there, we can find the approximation $\phi \approx \epsilon \approx \frac{1}{2}B$. Substituting accordingly gives:

$$R \approx 1 - \frac{1}{2}B$$

Looking at the expansion of decaying exponential:

$$e^{-x} \approx 1 - x$$

We can then deduce our approximation from there:

$$R \approx e^{-B/2}$$

5 Discrepancies of pole angles and peak frequencies

5.1 For what values of R and θ is the discrepancy greatest between the true peak frequency ω_m and the pole angle θ?

Using the equation derived in class for the true frequency:

$$\cos\omega_m = \frac{1+R^2}{2R}\cos\theta$$

We can see that as $R \to 0$ The discrepancy increases. This is because as the pole moves closer to the origin of the complex circle, the effect it has distorts the filter further.

Another feature that can be distinguished is that as $\theta \to 0, \pi$ the discrepancy increases because R shifts it away from the value of 1 or -1 respectively.

5.2 When θ is small, is ω_m shifted to higher or lower frequencies?

If $\theta < \frac{\pi}{2}$ then ω_m gets shifted to smaller frequencies. On the other hand, if $\theta > \frac{pi}{2}$ then ω_m gets shifted to larger frequencies. The reason that this occurs is that $\frac{\pi}{2}$ is the equilibrium point while each side is pulling towards either 0 or π.